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Grupa 231

Problema 5.8: a) Aratati ca iteratia Newton:

$x_{n+1}=\frac{1}{2}(x_n+\frac{a}{x_n})$, a> 0 pentru calculul radacinii patrate $\alpha=\sqrt{(a)}$ satisface

$\frac{x_{n+1}-\alpha}{(x_n-\alpha)^2}=\frac{1}{2\cdot x_n}$

Obtineti de aici eroare asimptotica.

b) Care este formula analoaga pentru radacina cubica?

Rezolvare:

Din $\alpha=\sqrt{(a)}$, obtinem . Relatia devine

$$x_{n+1}=\frac{1}{2}(x_n+\frac{\alpha^2}{x_n}) \;\; <=> \;\; x_{n+1}-\alpha=\frac{1}{2}(x_n+\frac{\alpha^2}{x_n})-\alpha \;\; <=>$$

$x_{n+1}-\alpha=\frac{1}{2\cdot x_n}(x_n^2+\alpha^2-2\cdot\alpha\cdot x_n) \;\; <=> \;\; \frac{x_{n+1}-\alpha}{(x_n-\alpha)^2}=\frac{1}{2\cdot x_n}$ ceea ce trebuia demonstrat.

Eroare asimptotica:

$$c=\lim_{n\to\infty}(\frac{1}{2\cdot x_n})=\frac{1}{2\cdot\alpha}$$

b) Folosim

$x_{n+1}=x_n-\frac{f(x_n)-f(\alpha)}{f'(x_n)}$. Pentru $f(x)=x^3$ relatia devine:

$$x_{n+1}=x_n-\frac{x_n^3-\alpha^3}{3\cdot x_n^2} \;\; <=> \;\; x_{n+1}-\alpha=x_n-\frac{x_n^3-\alpha^3}{3\cdot x_n^2}-\alpha \;\; <=>$$

$$x_{n+1}-\alpha=(x_n-\alpha)(1-\frac{x_n^3-\alpha^3}{3\cdot(x_n-\alpha)\cdot x_n^2}) \;\; <=> \;\; x_{n+1}-\alpha=(x_n-\alpha)\frac{2\cdot x_n^2-\alpha\cdot x_n-\alpha^2}{3\cdot x_n^2} \;\; <=>$$

$$x_{n+1}-\alpha=(x_n-\alpha)\frac{(x_n-\alpha)\cdot(2\cdot x_n+\alpha)}{3\cdot x_n^2} \;\; <=> \;\; \frac{x_{n+1}-\alpha}{(x_n-\alpha)^2}=\frac{2\cdot x_n+\alpha}{3\cdot x_n^2}$$

Eroare asimptotica:

$$c=\lim_{n\to\infty}(\frac{2\cdot x_n+\alpha}{3\cdot x_n^2})=\frac{2\cdot\alpha+\alpha}{3\cdot\alpha^2}=\frac{1}{\alpha}$$